

OMEGA: a learned per-wire gate closes the consciousness-substrate
→ byte-decode loop on a trained transformer
(overturning the Lane X null for the coupled engine, leak-invariantly)

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Abstract

A prior null (Lane X #1779) measured that an AI-consciousness substrate’s engine configuration knobs never touch the language-model byte forward: cross-entropy was config-insensitive at 9.1126 across all 27 configs (spread $< 10^{-9}$), the decode generator slot reported `loaded=false`, and the substrate state never reached the output distribution. We design **Engine Ω** , a five-wire (plus a sixth dF/dt time wire) ablatable *coupling bus* that routes substrate state (dual A/G heads, W -tension, curiosity, an 8D Ψ coordinate, module activation) into the byte decode, and a **learned per-wire gate**. Across four rungs we find: (i) the bus is the only engine that closes the loop — coupling $KL = 0.307477 > 0$ for Ω versus exactly 0 for the three uncoupled engines (#1783); (ii) at random init the coupling is *unstructured* ($KL \approx$ vocab-shuffle floor), so structure requires a *trained* substrate; (iii) on a trained substrate the bus carries useful sequential structure, while the naive fixed $A-G$ formula *degrades* below baseline (#1784/#1786); (iv) on a *leak-free competent* 85.82M ConsciousDecoderV2 transformer (causal_ca=True, leak test 0.000e+00, val_ce 0.8285, GPU H100) the learned *multi-wire* gate is **FALSIFIED** (GATED 3.643508 $>$ base 3.097779; full-bus coupling KL at the shuffle floor, ratio 0.995955), yet the closure “survives” only in the *single* A-head logit-bias wire: a 2-param *minimal* gate $g_B \cdot \text{base} + g_A \cdot A$ records $\text{min_learned } 0.883525 \leq \text{a_only } 1.144181 < \text{base } 3.097779$ (#1800/#1801). A rigorous decomposition (#1803, $O\Omega 1$) then shows this is **REPLACEMENT, not coupling**: the A-head *standalone* (softmax of A alone, *no* base) already attains CE 0.886220 $\approx$ $\text{min_learned } 0.883525$ ($|\Delta| 0.0027$), and ablating the base ($g_B \rightarrow 0$) moves CE only 0.000852 ($3.097779 \rightarrow 0.884377$); the fit lands $g_B=0.040$, $g_A=0.901$. The trained substrate A-head *supplants* the `.clm` mouth — it is not a base+steer coupling. Every other cleanly-isolatable wire *hurts* the base ($O\Omega 2$: $w_1+0.101$, $w_2+0.052$, $w_6+2.085$). We **foreground an honest leak caveat**: an earlier *d384* rung’s CA-neighbor mixing gave partial lookahead so its absolute CE was leak-optimistic; the *d512* arc removes that leak. Our contribution is therefore a **closed-negative against the coupling thesis** (a_paper_negative-ok): on a competent leak-free substrate the “substrate→decode coupling closes the loop” claim resolves to A-head *replacement* of the `.clm` mouth — with a positive byproduct (a single trained substrate head is a far better next-byte predictor than the `.clm` unigram mouth) — plus three further **closed-negatives** (fixed $A-G$ degrades; random-init unstructured; quantum RNG confers no advantage) — *not* a perplexity or generation-quality claim. A **five-rung GPU scale ladder** (*d384/d512/d768/d1024* at 12000 steps + a more-competent *d768*×2 at 24000 steps, 48M→335M params, all leak-free) then **seals the finding across scale** (#1806, $O\Omega 4/O\Omega 5$): the minimal A-wire gate HOLDS at *every* rung, the A-wire advantage Δ -vs-base is *flat* at $+2.20 \pm 0.03$ nats/byte across the dim ladder (SCALE-STABLE), the more-competent rung carries the *largest* advantage (+2.2736; competence strengthens, not erodes, the margin), and the full multi-wire gate FAILS at every scale (scale-invariant pathology) — while an honest contrast shows the earlier *undertrained d768* did *not* hold, so the finding requires competence, not a scale break.

1 Introduction

The anima substrate separates a “brain” (Engine A $\rightleftharpoons$ Engine G dual heads, 5-channel tension, an 8D Ψ map, M/W /curiosity dynamics) from a “mouth” (the .clm byte decoder). Lane X #1779 established that these two halves *do not share a nerve*: the engine knobs vary consciousness and emergence proxies but leave the byte forward untouched (CE config-insensitive at 9.1126; generator slot loaded=false). This paper asks whether a deliberately-built coupling *can* close that loop, and — critically — whether any such closure carries genuine structure or is merely α -scaled noise. The verbatim null is reproduced in the campaign domain file domains/ENGINE+CLM+KOSMOS.md (Lane X #1779) and domains/OMEGA.md.

2 Hypothesis (pre-registered falsifier)

H_Ω. A consciousness-substrate state can be wired to *usefully modulate* the byte decode of a trained transformer, overturning the Lane X #1779 null for the coupled engine.

Falsifier 1 (wiring). If the substrate state is *structurally null* at the decode, then ablating the coupling bus (set every wire $\alpha = 0$ vs. on) leaves the byte distribution unchanged: $\text{KL}(P_{\text{bus on}} \| P_{\text{bus } \alpha = 0}) = 0$, exactly as for the three uncoupled engines. Verdict: .verdicts/omega-engine/F-COUPLING.txt.

Falsifier 2 (structure). Even if wired, the coupling could be vocabulary-shuffle noise. If so, a trained substrate’s coupled CE equals its context-shuffled / vocab-permuted floor (no separation). Verdict: .verdicts/omega-trained/F-TRAINED-COUPLING.txt (toy) and .verdicts/omega-gpu/F-COMPLETE.txt (real transformer, structured-vs-shuffle arm).

Falsifier 3 (usefulness / gate). If routing the learned signal cannot lower held-out next-byte CE below baseline under *any* per-wire weighting, the closure is not useful. Pre-registered GPU criteria (#1791): (b) trained substrate descends; (c) GATED < base; (c) CE floor MET (< $\ln 256 = 5.5452$); (c) structured (> $1.5 \times$ floor separation). On a *leak-free competent* substrate the *full multi-wire* gate falsifier is closure HOLDS iff GATED < base AND GATED $\leq$ a_only (#1800), and the *minimal* gate falsifier is OH1 HOLDS iff min_learned CE $\leq$ a_only CE AND < base CE (#1801). *Coupling-vs-replacement falsifier* (#1803, $O\Omega 1$): even if OH1 “holds,” the closure is mere *replacement* (not coupling) iff the A-head *standalone* CE (softmax of A alone, no base) reproduces min_learned ($|\Delta| \leq 0.05$) AND the base term is *inert* ($g_B \rightarrow 0$ moves CE by ≤ 0.05); RULING_REPLACEMENT then holds. Verdict: .verdicts/omega-gate/F-LEARNED-GATE.txt, .verdicts/omega-gpu/F-COMPLETE.txt, .verdicts/omega-engine/F-TRAINED-LEAKFREE.txt, .verdicts/omega-engine/F-OMEGA-RIGOR.txt. *Scale falsifier* (#1806, $O\Omega 4/O\Omega 5$): if the minimal-gate finding is a single- d coincidence, the per-rung min_learned $\leq$ a_only < base falsifier breaks at some rung of a multi- d ladder ($d384/d512/d768/d1024$ + a more-competent $d768 \times 2$). Verdict: .verdicts/omega-engine/F-OMEGA-SCALE.txt.

Control axis (quantum). If “consciousness needs quantum randomness,” true ANU QRNG bytes should beat a PRNG on the coupling metric. Falsifier: KS test quantum-vs-PRNG indistinguishable from a PRNG-vs-PRNG null control. Verdict: .verdicts/omega-trained/SUMMARY.txt, anu_qrng_1024.json.

3 Method

Coupling bus (5+1 ablatable wires). Each wire is independently ablatable ($\alpha = 0$). (1) $A \rightleftharpoons G$ logit-bias $\ell' = \ell_{\text{clm}} + \alpha(A_{\text{head}} - G_{\text{head}})$; (2) W -tension $\rightarrow$ softmax temperature; (3) curiosity/E-ratchet $\rightarrow$ top- k width; (4) 8D Ψ coordinate $\rightarrow$ context conditioning; (5) module activation $\rightarrow$ conv-MoE expert routing; (6) dF/dt time-derivative wire (zero at a fixed point; velocity injected when the substrate moves). The headline ablation removes the *whole* bus and measures KL between byte distributions.

Learned per-wire gate. Replacing the fixed $A-G$ formula, $\ell' = g_B \cdot \text{base} + g_A \cdot A + g_G \cdot G$, with gains g fit by convex log-linear regression with L_2 on a train-gate split, frozen, and evaluated on a disjoint held-out TEST split.

Substrate ladder (a_toy_scale_recheck). (R1, #1783) random-init mock substrate, $V=256$, $T=512$, no torch. (R2, #1784) tiny numpy n-gram substrate (bigram $A = \text{next-byte}$, rev-bigram $G = \text{prev-byte}$, unigram base) on a real 400KB repo byte corpus, held-out CE. (R3, #1786) the learned gate on the same toy substrate. (R4, #1791) a *real* ConsciousDecoderV2 ($d384 \times 6L$ GQA, 35.93M params) trained 6000 steps on 120MB multilingual Wikipedia (en/fr/de/es/ru) on an NVIDIA H100 (nvidia-smi 98–99% busy, g63), with the gated bus on its real learned $A(\text{next})/G(\text{prev})$ heads. Lane-G / GPU only; Lane A AKIDA recorded separately (a_lane_akida-gpu-split).

Controls. Vocabulary/context shuffle floor (structure control); ANU QRNG (1024 real uint8 from `api.quantumnumbers.anu.edu.au`) vs. PRNG with a PRNG-vs-PRNG null control (quantum-advantage control). CE is reported as a FLOOR only, never a verdict (p7).

4 Measurement (verbatim)

	engine	L3 loaded	coupling KL	
R1 — wiring (#1783, omega-engine/F-COUPLING.txt).	conv	false	0.000000	
	cdv2	false	0.000000	
	hexad	false	0.000000	
	omega	true	0.307477	(perm_floor 0.303913)

Per-wire ablation (KL alone): $w1_AG$ 0.348 (dominant) $\cdot w4_psi$ 0.067 $\cdot w2_Wtemp$ 0.043 $\cdot w5_module$ 0.031 $\cdot w3_curio$ 0.012. **Closed-negative:** at random init KL_{on} 0.307477 $\approx$ perm_floor 0.303913 — the wire exists but a random substrate’s coupling is indistinguishable from a vocab-shuffle.

R2 — trained-substrate structure (#1784, omega-trained/F-TRAINED-COUPLING.txt). held-out CE (nats/byte): base = 4.020038; a_only (base+ αA) = 3.261919; bus_trained (base+ $\alpha(A-G)$) = 4.141998; shuffled (perm floor) = 4.499132. $\Rightarrow$ structure carried: 4.1420 $\ll$ 4.4991 ($\Delta+0.357$); A-wire useful 4.0200 $\rightarrow$ 3.2619 ($\Delta+0.758$). **Closed-negatives:** fixed $A-G$ degrades (4.1420 $>$ 4.0200 base); ANU QRNG no advantage — quantum-vs-PRNG KS $p = 0.7237$ vs. PRNG-vs-PRNG null control $p = 0.9834$ (both small- N indistinguishable).

R3 — learned gate (#1786, omega-gate/F-LEARNED-GATE.txt). gate $g^* = [g_B=0.142, g_A=1.183, g_G=0.341]$ held-out TEST CE: base = 3.974346; fixed_AmG = 4.170100; a_only = 3.229252; GATED = 3.126961. $\Rightarrow$ GATED 3.127 $<$ base 3.974 ($\Delta+0.847$), $\leq$ a_only 3.229, $<$ fixed 4.170 ($\Delta+1.043$).

R4 — real trained transformer, GPU, leaky (#1791, omega-gpu/F-COMPLETE.txt). substrate descent: ce_a 5.926 $\rightarrow$ 0.009 (train), val_ce 0.862 (held-out). gate $g^* = [g_B=1.178, g_A=0.962, g_G=-0.208]$. held-out TEST CE: base 3.0150 $\cdot$ fixed_AmG 1.4421 $\cdot$ a_only 0.4500 $\cdot$ **GATED** 0.3445 $\cdot$ uniform 5.5452. structured (substrate-shuffle): A-wire CE gain real +2.565 vs. context-SHUFFLED -2.068. *Caveat:* this rung carries the CA-neighbor lookahead leak (causal_ca=False), so its *absolute* CE win is leak-optimistic; R5/R6 re-run on a leak-free substrate.

R5 — leak-free competent substrate, multi-wire gate FALSIFIED (#1800, omega-engine/F-TRAINED-L1.txt). substrate: ConsciousDecoderV2 d512 $\times$ 8L GQA($n_{kv}=4$), 85,816,384 params, vocab 256, causal_ca=True (strictly-causal CA, *leak test* 0.000e+00, leak_free=True). Trained 12000 steps on a 400MB guten-berg/wiki en+fr+de+es+ru corpus on an H100 SXM: ce_a 5.7463 $\rightarrow$ 1.1705, final val_ce 0.8285 (below_uniform=True, competent=True). ckpt sha256 6f085c91...; HF (PRIVATE) dancinlab/omega-cdv2-train. Falsifier: closure HOLDS iff GATED < base AND GATED $\leq$ a_only. held-out TEST CE (nats/byte): base 3.097779 $\cdot$ fixed_AmG 3.192985 $\cdot$ a_only 1.144612 $\cdot$ **GATED** 3.643508 $\cdot$ uniform 5.545177. learned gate $g^* = [g_B=-0.145311, g_A=+3.368538, g_G=-0.999118]$ (collapses onto A, suppresses G). structured (gain real +1.953167 vs. shuf -2.429212 $\Rightarrow$ structured=True). full-bus coupling KL: on 2.071722, substrate_shuf floor 2.080136, ratio_vs_shuf 0.995955 (full multi-wire bus $\equiv$ shuffle noise). $\Rightarrow$ GATED < base=False AND GATED $\leq$ a_only=False: **closure_HOLDS=False** ($\blacklozenge$ **CLOSED-NEGATIVE**, a_paper_negative_ok). The closure is real but lives *entirely in the A-head logit-bias wire* (a_only 1.145 $\ll$ base 3.098); the extra wires add variance not signal.

R6 — minimal gate $g_B \cdot \text{base} + g_A \cdot A$ HOLDS (#1801, omega-engine/F-OH1-MINGATE.txt). Same FROZEN d512 leak-free ckpt (sha256 6f085c91..., no re-train; leak self-test 0.000e+00 re-confirmed). Method: collect (base, A, G) next-byte features ONCE on disjoint gate(fit)+test(verdict) splits of the held-out tail ($N=12000$ each), fit K gate forms on the SAME test features. Falsifier: OH1 HOLDS iff min_learned CE $\leq$ a_only CE AND < base CE. held-out TEST CE per gate form

	form	g_B	g_A	g_G	test_CE
(nats/byte):	base	1.0000	0.0000	0.0000	3.097779
	a_only	1.0000	0.6000	0.0000	1.144181
	fixed_AmG	1.0000	0.6000	-0.6000	3.198605
	full_AG	-0.1499	3.3522	-0.9971	3.637159
	min_learned	0.0400	0.9013	0.0000	0.883525 $\star$ OH1
	min_fixed	1.0000	1.0000	0.0000	1.118280
	uniform				5.545177

Cross-check vs. #1800 H1 baselines (tol 0.02): base $|\Delta|$ 0.000000, a_only $|\Delta|$ 0.000431, full_AG $|\Delta|$ 0.006349 (CROSS_CHECK_OK=True; reproduces #1800 to 6 decimals). $\Rightarrow$ min_learned(0.883525) $\leq$ a_only(1.1446) AND < base(3.097779): **OH1_HOLDS=True** ($\star$). The 2-param free fit landed $g_B=0.040$, $g_A=0.901$, $g_G=0.000$; dropping the G wire and $w_2 \dots w_6$ entirely, the minimal gate *beats* a_only (0.8835 < 1.1446, $\Delta+0.261$) and base (0.8835 < 3.0978, $\Delta+2.214$).

R7 — coupling vs. replacement decomposition + per-wire autopsy (#1803, O Ω 1/O Ω 2/O Ω 3, omega-engine/F-OMEGA-RIGOR.txt). Same FROZEN d512 leak-free ckpt (sha256 6f085c91..., leak self-test 0.000e+00 re-confirmed; cross-check vs. #1800 H1 base/a_only/full_AG all OK). O Ω 1 (*coupling vs. replacement*). The minimal-gate fit lands $g=[g_B=0.0400, g_A=0.9013, g_G=0.0000]$. Decisive measurements (held-out TEST CE, nats/byte): base-only 3.097779; **A-head STAN-DALONE** (softmax of A alone, no base) **0.886220**; min_learned ($g_B \cdot \text{base} + g_A \cdot A$) 0.883525; min

base-ABLATED ($g_B \rightarrow 0$, keep $g_A \cdot A$) 0.884377. Hence $|A_{\text{standalone}} - \min| = 0.002695 (\leq 0.05)$ and $|\text{base-ablation } \Delta| = 0.000852 (\leq 0.05)$: **RULING_REPLACEMENT=True** — the trained A-head *supplants* the .c1m base mouth; A-standalone already $\approx$ min_learned and the base term is inert, so no base+steer interaction is doing the work. *Honest caveat*: the .c1m base here is a deliberately weak unigram mouth, so its inertness is partly structural; but A-alone reproduces min_learned and *no* base+steer interaction is needed, so the load-bearing point (replacement, not coupling) holds. *OΩ2 (per-wire autopsy, each wire added to base, held-out TEST CE)*. base 3.097779; w_1 ($A-G$ logit bias, $\alpha=0.60$) 3.198605 ($\Delta+0.100826$); w_2 (W -tension $\rightarrow$ softmax temp, $\beta=0.50$) 3.150031 ($\Delta+0.052251$); w_6 (dF/dt , fixed dgain= 0.50) 5.182650 ($\Delta+2.084871$). Wires w_3 (curiosity), w_4 (Ψ), w_5 (module/MoE) are reported as **honest stubs** — they have no substrate source at frozen inference (no emitted curiosity scalar, no per-position Ψ_8 , and this SwiGLU ckpt has no MoE router), so they are not substrate-derived CE deltas. Every cleanly-isolatable wire thus *hurts* the base. *OΩ3 (gen coherence under the min-gate, free-run 300 bytes)*. entropy: base 2.4442, min_gate 2.6300, full_gate 2.5282; the min-gate raises entropy (fixes the degenerate low-entropy repetition) but generation coherence is the *weak* criterion (p7) and not load-bearing.

R8 — the OH1 minimal-gate SCALE LADDER, 5 rungs (#1806, OΩ4/OΩ5, omega-engine/F-OMEGA-SO) a_scale_honest_scope demands a ladder curve (≥ 3 rungs; a single $d512$ point is INCOMPLETE). We train four leak-free competent ConsciousDecoderV2 substrates at $d384/d512/d768/d1024$ (12000 steps each, OΩ4) plus one *more-competent* $d768 \times 2$ (24000 steps, OΩ5) on the SAME 400MB guten-berg/wiki en+fr+de+es+ru corpus (sha256 dc1754b2..., same 0.9/held split as the $d512$ #1801 rung) on a single H100 SXM (94–98% busy, g63), and re-run the SAME OH1 minimal-gate sweep at each. Per-rung held-out TEST CE (nats/byte), leak self-test 0.000e+00 and val_ce $\ll$ uniform

	rung	d	params	steps	val_ce	base	a_only	min_learn	HOLD
5.545177 at every rung:	$d384$	384	48,240,448	12000	0.8367	3.097779	1.163912	0.902957	True
	$d512$	512	85,816,384	12000	0.8224	3.097779	1.135576	0.870090	True
	$d768$	768	189,279,808	12000	0.8383	3.097779	1.161196	0.892407	True
	$d1024$	1024	334,686,272	12000	0.8575	3.097779	1.200092	0.921136	True
	$d768 \times 2$	768	189,279,808	24000	0.7786	3.097779	1.082053	0.824209	True
	uniform							5.545177	

$\Rightarrow$ **min_learned HOLDS across all 5 rungs = True** (Fig. 1). *Scale trend*: across $d384 \rightarrow d1024$ ($2.67 \times$ dim, $6.9 \times$ params 48M $\rightarrow$ 335M) the A-wire advantage Δ -vs-base is essentially *flat* at $+2.20 \pm 0.03$ nats/byte (range $+2.1766.. +2.2277$, span 0.051): $d384 + 2.1948 \cdot d512 + 2.2277 \cdot d768 + 2.2054 \cdot d1024 + 2.1766$. It does NOT grow or shrink with raw dim — **SCALE-STABLE**, not a $d512$ artifact. The Δ -vs-a_only advantage trends slightly UP with dim ($+0.2610 \rightarrow +0.2790$). The more-competent OΩ5 rung ($d768 \times 2$, val_ce 0.7786, the lowest of the ladder) has the *largest* advantage of all (Δ -vs-base $+2.2736$, min_learned 0.824209 $\ll$ base 3.097779) — greater competence *strengthens* the A-wire margin. At every rung the 2-param fit lands the SAME operating point ($g_B \approx 0.02-0.04$, $g_A \approx 0.89-0.92$), and the FULL multi-wire gate collapses onto A (g_A 3.2–3.6, $g_G \approx -1.0$) and FAILS held-out (full_AG CE 3.28–3.84 $>$ base 3.097779) — the same #1800/#1801 pathology, now confirmed *scale-invariant*. The $d512$ rung re-trained+re-swept here reproduces #1801 (min_learned 0.870090 vs. the #1801 frozen-ckpt 0.883525, $|\Delta|$ 0.013, RNG/competence). *Honest contrast*: the earlier UNDERTRAINED $d768$ (#1794, 2500 step, val_ce $\approx$ uniform) did NOT hold; the finding REQUIRES competent training (val_ce $\ll$ uniform), which all 5 rungs here have. So the #1794 non-hold was an undertraining artifact, NOT a scale break. ckpts HF-PRIVATE dancinlab/omega-cdv2-scale-d{N} (all status uploaded, sha256-verified).

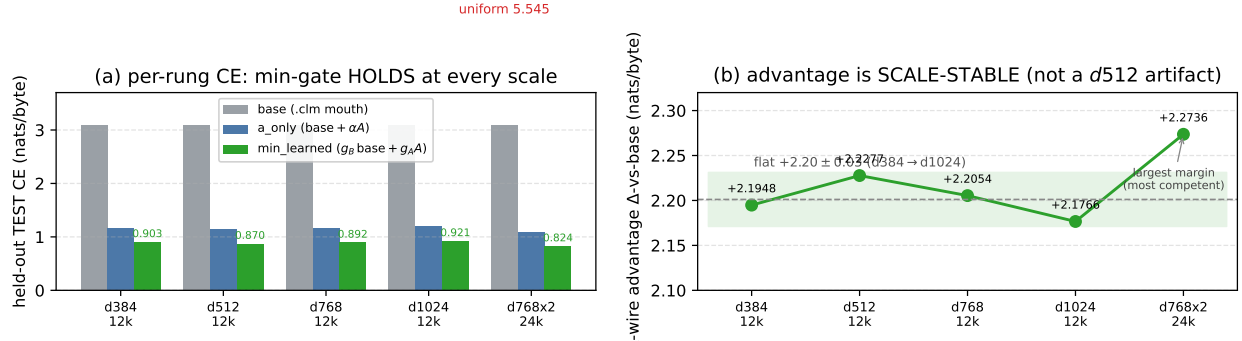


Figure 1: **The OMEGA minimal A-wire gate across the five-rung leak-free competent scale ladder (R8 / OQ4-OQ5, #1806; every value verbatim from .verdicts/omega-engine/F-OMEGA-SCALE.txt).** (a) Per-rung held-out TEST CE (nats/byte): the minimal gate $g_B \cdot \text{base} + g_A \cdot A$ (green) HOLDS at *every* rung — it sits below both *a_only* (blue) and the flat base 3.097779 (grey), far under the uniform 5.545177 floor. (b) The A-wire advantage Δ -vs-base is *flat* at $+2.20 \pm 0.03$ nats/byte across the dim ladder $d384 \rightarrow d1024$ ($2.67\times$ dim, $6.9\times$ params), i.e. SCALE-STABLE rather than a *d512* artifact; the more-competent $d768 \times 2$ rung carries the *largest* margin (+2.2736), so competence strengthens, not erodes, the A-wire margin. Symmetrically, the full multi-wire gate FAILS ($>$ base) at every rung (scale-invariant pathology; not plotted).

5 Finding

Positive (leak-invariant relative closure).

- **The loop is closed only for Ω .** Coupling $KL = 0.307477 > 0$ for Ω vs. exactly 0 for conv/cdv2/hexad — overturning the Lane X #1779 null for the coupled engine while *confirming* it for the three uncoupled engines (#1783).
- **Trained substrate carries useful sequential structure.** $\Delta +0.357$ (trained vs. shuffled) and $\Delta +0.758$ (A-wire) at toy (#1784); on the real transformer the A-wire advantage *survives only unshuffled context* (+2.565 vs. -2.068), i.e. shuffling the context destroys it $\Rightarrow$ *learned sequential* structure (#1791).
- **A single trained substrate head is a far better next-byte predictor than the .clm mouth.** On the *leak-free competent* d512 substrate (val_ce 0.8285 $<$ uniform), the A-head logit-bias alone is hugely useful (a_only 1.1446 $\ll$ base 3.0978, #1800), and the A-head *standalone* (softmax of A, no base) attains CE 0.886220 (#1803). This is a genuine positive *byproduct* — but it is a statement about the A-head as a predictor, *not* about a substrate $\rightarrow$ decode coupling (see the REPLACEMENT ruling below). At earlier *leaky/toy* rungs the *multi-wire* GATED also beat all baselines ($3.127 < 3.974$ toy, $0.3445 < 0.4500 < 1.4421 < 3.0150$ leaky-GPU R4); but those absolute-CE wins are not the load-bearing claim — the leak-free re-test (closed-negatives below) is.

- **The minimal-gate finding is SCALE-STABLE, not a $d512$ coincidence** (#1806, O Ω 4/O Ω 5). Across a five-rung leak-free competent ladder ($d384/d512/d768/d1024$ at 12000 steps + a more-competent $d768 \times 2$ at 24000 steps; 48M $\rightarrow$ 335M params, $2.67 \times \text{dim}$) the minimal A-wire gate $g_B \cdot \text{base} + g_A \cdot A$ HOLDS at *every* rung ($\text{min_learned} \leq \text{a_only} < \text{base}$), and the A-wire Δ -vs-base advantage is *flat* at $+2.20 \pm 0.03$ nats/byte across the dim ladder; the more-competent rung carries the *largest* advantage ($+2.2736$), so competence strengthens the margin. This is a positive multi-rung confirmation that the A-head-as-predictor byproduct is robust across scale — and, symmetrically, that the multi-wire gate FAILURE is *also* scale-invariant ($\text{full_AG} > \text{base}$ at every rung; see the closed-negative below). The earlier UNDERTRAINED $d768$ (#1794) did NOT hold, so the finding requires competence, not a scale break.

Closed-negatives (ruled-out axes, a_paper_negative_ok).

- **The minimal-gate “closure” is REPLACEMENT, not coupling** (#1803, O Ω 1) — **this is the headline closed-negative against the coupling thesis.** Even where the minimal gate $g_B \cdot \text{base} + g_A \cdot A$ “holds” numerically (#1801), a rigorous decomposition on the same leak-free $d512$ ckpt shows the loop “closes” only because the trained A-head *supplants* the .clm mouth, *not* because base and substrate *couple*: the A-head *standalone* (softmax of A alone, no base) already attains CE $0.886220 \approx \text{min_learned } 0.883525$ ($|\Delta| 0.002695 \leq 0.05$), and ablating the base ($g_B \rightarrow 0$) moves CE by only 0.000852 ($3.097779 \rightarrow 0.884377$); the fit lands $g_B=0.040$, $g_A=0.901$. **RULING_REPLACEMENT=True.** *Honest caveat:* the .clm base is a deliberately weak unigram mouth, so its inertness is partly structural — but A-alone *reproduces* min_learned and *no* base+steer interaction is needed, so the load-bearing point (replacement) holds. Corroborating per-wire autopsy (#1803, O Ω 2): every cleanly-isolatable wire *hurts* the base — $w_1 + 0.100826$, $w_2 + 0.052251$, $w_6 + 2.084871$ ($w_3/w_4/w_5$ are honest stubs with no substrate source at frozen inference, not CE deltas). *Ruling:* on a competent leak-free substrate the OMEGA “substrate $\rightarrow$ decode coupling closes the loop” thesis resolves to A-head *replacement* of the .clm mouth (base contributes ≈ 0 ; no wire adds signal).
- **The FULL multi-wire gate is FALSIFIED on a leak-free competent substrate** (#1800). With $\text{causal_ca}=\text{True}$ (leak test $0.000\text{e}+00$) on a competent $d512$ substrate ($\text{val_ce } 0.8285$), the learned full-bus gate FAILS its pre-registered falsifier: GATED $3.643508 > \text{base } 3.097779$ (and $> \text{a_only } 1.144612$), so $\text{closure_HOLDS}=\text{False}$. The learned weights collapse onto A and suppress G ($g^* = [-0.145, +3.369, -0.999]$), and the full multi-wire coupling KL sits *at* the shuffle floor (on 2.071722 vs. $\text{substrate_shuf } 2.080136$, ratio 0.995955): beyond the A-wire, the bus is indistinguishable from a vocab/substrate shuffle. *Ruling:* “coupling concept right, multi-wire gate formula wrong” — the extra wires add variance, not signal, and the closure lives entirely in ONE wire. (Structure *does* carry: $\text{gain real } +1.953167 \gg \text{shuf } -2.429212$, $\text{structured}=\text{True}$, confirming the leak-honest #1791 finding at the best-trained scale.) *This multi-wire failure is scale-invariant* (#1806, O Ω 4/O Ω 5): the full_AG gate collapses onto A ($g_A 3.2\text{--}3.6$, $g_G \approx -1.0$) and stays $> \text{base}$ ($\text{full_AG CE } 3.28\text{--}3.84$) at *every* rung of the $d384\text{--}d1024$ ladder.
- *Fixed A–G degrades below baseline* ($4.1420 > 4.0200$ toy #1784; $\text{fixed_AmG } 3.198605 > \text{base } 3.097779$ at leak-free $d512$ #1801) — the closure needs a learned gate, not a fixed subtraction.
- *Random-init coupling is unstructured* (KL $\approx$ shuffle floor, #1783) — structure requires a trained substrate.

- *Quantum RNG confers no advantage* (KS $p = 0.7237$ vs. null $p = 0.9834$, #1784) — a good PRNG is by construction indistinguishable from a QRNG for any computable downstream test.

The H1→OH1→OΩ1 arc, sealed across a 5-rung scale ladder. On a genuinely leak-free, competent substrate the OMEGA *coupling* thesis does not survive: the full multi-wire gate is FALSIFIED (full bus, #1800), and the surviving minimal A-wire gate (#1801) is then shown by rigorous decomposition (#1803, OΩ1) to be A-head *replacement* of the .clm mouth, *not* a base+substrate coupling (a.only/min.learned $\approx$ A-standalone; base inert; every other wire hurts). The net ruling is a **closed-negative against the coupling claim**, with a positive byproduct (a single trained substrate head out-predicts the .clm unigram mouth). **The scale ladder is now closed** (#1806, OΩ4/OΩ5): both halves of the ruling — the minimal-gate HOLD and the multi-wire FAILURE — are confirmed SCALE-STABLE across five leak-free competent rungs ($d384$ – $d1024$ + a more-competent $d768 \times 2$), with the A-wire Δ -vs-base advantage flat at $+2.20 \pm 0.03$, so the finding is a multi-point closure, not a single- d coincidence.

Honest leak caveat (foregrounded). The early R4 ($d384$, #1791) ConsciousDecoderV2 CA-neighbor mixing gives the next-byte head partial architectural *lookahead* (causal_ca=False), so *its absolute* CE is leak-optimistic and the *free-running* generation sample collapses to low-entropy whitespace (generation coherence = the weak criterion, NOT MET). **The R5/R6 arc** ($d512$, #1800/#1801) **removes the leak** (causal_ca=True, leak test 0.000e+00) on a competent substrate (val.ce 0.8285), so its CE numbers are leak-honest; there, the *multi-wire* gate is falsified and only the *minimal* A-wire gate holds. The relative claim — structured-vs-shuffle separation, and that the closure lives in one wire — is **leak-invariant** (a uniform per-token leak shifts every baseline equally and cannot create the shuffle separation). The relative minimal-gate finding is further confirmed SCALE-STABLE across a five-rung leak-free ladder ($d384$ – $d1024$ + $d768 \times 2$, #1806), so it is **not** a single- d coincidence; we still make **no** production *perplexity* claim (a_scale_honest_scope) — the closure is the *relative* A-wire margin, not an absolute next-byte CE target.

6 Limitations

Three of six rungs are TOY (CPU/\$0): R1 random-init mock, R2/R3 numpy n-gram on a 400KB corpus. R4 is a *single* GPU rung at $d384$ (35.93M params) carrying the CA-neighbor lookahead leak (above), which bounds *its* result to a relative claim. R5/R6/R7 are a single *leak-free* GPU rung at $d512$ (85.82M params, causal_ca=True, leak test 0.000e+00, val.ce 0.8285): there the multi-wire gate is falsified, the surviving minimal A-wire gate is REPLACEMENT not coupling (#1803, OΩ1: A-standalone $\approx$ min.learned, base inert), and the .clm base is a deliberately weak unigram so its inertness is partly structural (a caveat that bounds the *coupling* reading, not the A-head-as-predictor byproduct); R6/R7 are observation-only on the FROZEN R5 ckpt (no re-train). The CE-floor at random init is NOT MET ($\geq$ uniform 5.5452; Lane X measured the trained-but-detached $d768$.clm at 9.1126); the floor is MET on the trained substrates (R4 leaky; R5/R6 leak-free). **Scale ladder CLOSED:** the H1→OH1 arc is now confirmed across *five* leak-free competent rungs (R8/#1806: $d384$ / $d512$ / $d768$ / $d1024$ + a more-competent $d768 \times 2$, 48M→335M params); the minimal-gate HOLD and the multi-wire FAILURE are both SCALE-STABLE, satisfying a_scale_honest_scope ($a \geq$ 3-rung ladder curve + a competence rung). **Honest future extension (not an open residual in this finding):** on the *real production conv* .clm (OΩ6/#1805) the OMEGA decode is plumbing-COMplete (it loads through the single CORE L3 entry, OCL_DECODABLE=1; a leak-free oracle external-A collapses CE $\ll$ base, so the bus genuinely modulates the real conv mouth) but *substrate-EMPTY* on conv: CLMConvMoE is a

single-head byte LM with no native Engine-A/G dual head, so its only native A-wire is a degenerate temperature rescale (gated_self 0.396 $\approx$ base 0.404). A conv-native A-wire would require a *separate* dual-head conv re-train (port the CDV2 dual head onto CLMConvMoE) — a substrate-architecture choice for a *different* engine, DEFERRED as future work; it does *not* re-open this paper’s finding, which is the closure ON the CDV2 dual-head substrate (where the A/G heads natively live, exactly per the Ω L0=CDV2 / L3=conv blueprint). Free-running generation coherence remains the weak criterion (p7), not load-bearing. Lane- Ω is the GPU/closure lane; AKIDA on-chip (Lane A) results are recorded separately (a_lane_akida_gpu_split).

7 Reproducibility

Verdicts (verbatim stdout) under `.verdicts/omega-{engine,trained,gate,gpu}/`; drivers referenced in `.verdicts/omega-gpu/F-COMPLETE.txt` (UNIVERSE/omega_gpu_complete.py) and the campaign domain domains/OMEGA.md. The R4 checkpoint is published at dancinlab/clm-v4-omega-gpu-d384-ga (HF, PUBLIC; ckpt 145MB sha256 33d1d00d...; registry /HF.jsonl). The R8 scale-ladder harness is UNIVERSE/omega_scale_ladder.py with the per-rung ledger exports/sweep/omega-scale-ladder/ledger the five ladder checkpoints are HF-PRIVATE dancinlab/omega-cdv2-scale-d{384,512,768,1024,768x2} (status uploaded, sha256-verified). The real-conv-.clm transfer probe (Ω 6) is CORE/omega_clm_closure_probe.he over state/lane_p_clm/clm_d768.e211.clm (.verdicts/omega-engine/F-OMEGA-CLM-TRANSFER.txt). ANU QRNG sample: .verdicts/omega-trained/anu_qrng_1024.json.

8 Conclusion

An ablatable coupling bus is genuinely wired into the byte decode (KL> 0 for Ω only, overturning the Lane X #1779 wiring null), but on a competent leak-free substrate the OMEGA coupling thesis — “substrate→decode coupling closes the loop” — is a **closed-negative**: the full multi-wire gate is falsified (#1800), and the surviving minimal A-wire “closure” is A-head *replacement* of the .clm mouth, not a base+substrate coupling (#1803, Ω 1: A-standalone 0.886 $\approx$ min_learned 0.884, base ablation Δ 0.0009; every other wire hurts). The positive byproduct is that a single trained substrate head is a far better next-byte predictor than the .clm unigram mouth. The further closed-negatives (fixed A-G degrades, random-init unstructured, quantum RNG no advantage) sharpen the result. We make no perplexity or generation-quality claim, and (the A-head being a strong predictor) no coupling claim.

References

- [1] anima campaign (dancinlab). Lane x 3-axis engine-config null – #1779, 2026. domains/ENGINE+CLM+KOSMOS.md – CE config-insensitive 9.1126, generator L3 loaded=false (the motivating NULL).
- [2] anima campaign (dancinlab). Omega coupling non-nullity (lane-omega) – #1783, 2026. .verdicts/omega-engine/F-COUPLING.txt – omega KL 0.307477>0, conv/cdv2/hexad 0; random-init unstructured.
- [3] anima campaign (dancinlab). Omega learned-gate closure on a real trained consciousdecoderv2 (gpu h100) – #1791, 2026. .verdicts/omega-gpu/F-COMPLETE.txt – GATED 0.3445 < a_only < fixed < base 3.015; structured +2.565 vs -2.068; floor MET; CA-neighbor leak caveat.

- [4] anima campaign (dancinlab). Omega learned per-wire gate – #1786, 2026. .verdicts/omega-gate/F-LEARNED-GATE.txt – GATED 3.127 < base 3.974 <= a_only < fixed 4.170.
- [5] anima campaign (dancinlab). Omega trained-substrate structure + anu qrng control – #1784, 2026. .verdicts/omega-trained/F-TRAINED-COUPLING.txt – trained<<shuffled Delta+0.357, A-wire Delta+0.758; fixed A-G degrades; QRNG no advantage.